

Quantum Mechanics — Notes (Qm9)

Spin states, addition of angular momenta (Lec 23), problem set, and Lec 24

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1 Spin states of $\vec{S} \cdot \hat{n}$ (contd.)

1.1 Normalized spin state

$$a^2 \left(1 + \tan^2 \frac{\theta}{2} e^{2i\phi}\right) = 1 \Rightarrow a^2 = \cos^2 \frac{\theta}{2} \Rightarrow a = \pm \cos \frac{\theta}{2}$$

$$|\nearrow\rangle = \begin{pmatrix} \cos \theta/2 \\ e^{i\phi} \sin \theta/2 \end{pmatrix} \quad \dots \text{normalized state (How)?}$$

“So, If you give me any arbitrary $|ket\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$ such that $|a|^2 + |b|^2 = 1$, we could interpret that $|\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$ as eigen state of $\hat{S} \cdot \hat{n}$ with an appropriate $\hat{n}(\theta, \phi)$. The polar angle are reflected as $\begin{pmatrix} \cos \theta \\ e^{i\phi} \sin \theta \end{pmatrix}$.”

Case (1) if $\theta = 0$, $\Rightarrow$ axis of quantization is $\hat{z}$ axis

$$|\chi\rangle = \begin{pmatrix} \cos 0 \\ e^{i\phi} \sin 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\theta = \pi \hat{z}; \quad |\chi\rangle = \begin{pmatrix} \cos \frac{\pi}{2} \\ e^{i\phi} \sin \frac{\pi}{2} \end{pmatrix} = + \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$\hookrightarrow$ eigen states $|S_z = \pm \frac{\hbar}{2}\rangle$

1.2 (2) Eigen state of $\hat{S}_x$

(2) find eigen state of $\hat{S}_x$ if its eigen value is $\frac{\hbar}{2}$.

$|S_x = +\frac{\hbar}{2}\rangle$ (in $\hat{n}$ axis of quantization; which is x axis).

what kind of states corresponds to $+\frac{\hbar}{2}$ eigen value

+ve x -axis $\theta = \pi/2, \phi = 0, 2\pi \dots$

$$|\chi\rangle = \begin{pmatrix} \cos \pi/4 \\ e^{i0} \sin \pi/4 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$

$$\& \quad |S_x = -\frac{\hbar}{2}\rangle = \begin{pmatrix} \cos(-\pi/4) \\ e^{i0} \sin(-\pi/4) \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$|S_x = \pm \frac{\hbar}{2}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm 1 \end{pmatrix}$$

1.3 (3) Eigen states of $\hat{S}_y$

(3) find $|S_y = \pm \frac{\hbar}{2}\rangle$ (eigen states of $\vec{S} \cdot \hat{j} = \hat{S}_y$). ($\hat{n} = \hat{j}$ here)

$$|\chi\rangle = \begin{pmatrix} \cos \theta/2 \\ e^{i\phi} \sin \theta/2 \end{pmatrix} \quad \begin{array}{l} \theta = \pi/2 \\ \phi = \pi/2, 3\pi/2 \end{array}$$

$$|\chi\rangle = \begin{pmatrix} \cos \pi/4 \\ e^{i\pi/2} \sin \pi/4 \end{pmatrix} \quad \text{or} \quad \begin{pmatrix} \cos \pi/4 \\ e^{i\frac{3\pi}{2}} \sin \pi/4 \end{pmatrix}$$

$$|\chi\rangle \Rightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ \pm i \end{pmatrix}$$

$$|S_y = +\frac{\hbar}{2}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |S_y = -\frac{\hbar}{2}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

1.4 (Q) Probability of measuring S_x eigen values

(Q) what if we measure probability of $\hat{S}_x$, what is probability of measuring $+\frac{\hbar}{2}$ or $-\frac{\hbar}{2}$ eigen values.

If we measure S_z , $P_{\uparrow} = |a|^2$, $P_{\downarrow} = |b|^2$ for $|\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$

but instead we measure $\hat{S}_x$, what is $P_{\rightarrow}$ and $P_{\leftarrow}$?

let us expand $|\chi\rangle$ in $|\rightarrow\rangle$ and $|\leftarrow\rangle$ basis

$$|\rightarrow\rangle_{S_x=+\frac{\hbar}{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ +1 \end{pmatrix}, \quad |\leftarrow\rangle_{S_x=-\frac{\hbar}{2}} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix}$$

$$\langle \rightarrow | \leftarrow \rangle = 0 \quad \left(\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = 0 \right)$$

so let us expand in this basis

$$|\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix} = c |\rightarrow\rangle + d |\leftarrow\rangle$$

$$c = \langle \rightarrow | \chi \rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{a}{\sqrt{2}} + \frac{b}{\sqrt{2}} = \frac{a+b}{\sqrt{2}}$$

$$d = \langle \leftarrow | \chi \rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} = \frac{a-b}{\sqrt{2}}$$

$$P_{|\rightarrow\rangle} = |c|^2 = \frac{(a+b)^2}{2}, \quad P_{|\leftarrow\rangle} = |d|^2 = \frac{(a-b)^2}{2}$$

1.5 (Q) Probability of measuring S_y eigen values

(Q) what are probabilities of measuring $S_y = \pm\hbar/2$ for any abstract spin state $|\chi\rangle = \begin{pmatrix} a \\ b \end{pmatrix}$.

let us expand $|\chi\rangle$ in basis of eigen states of $\hat{S}_y$

$$|S_y = \frac{\hbar}{2}\rangle = |\nearrow\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \quad |S_y = -\frac{\hbar}{2}\rangle = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$|\chi\rangle = c |\nearrow\rangle + d |\searrow\rangle$$

where,

$$c = \langle \nearrow | \chi \rangle = \frac{1}{\sqrt{2}} (1 \quad -i) \begin{pmatrix} a \\ b \end{pmatrix} = \frac{(a - ib)}{\sqrt{2}}$$

$$d = \langle \searrow | \chi \rangle = \frac{1}{\sqrt{2}} (1 \quad +i) \begin{pmatrix} a \\ b \end{pmatrix} = \frac{(a + ib)}{\sqrt{2}}$$

$$\left. \begin{aligned} P_{|\nearrow\rangle} &= |c|^2 = \frac{(a+ib)(a-ib)}{2} = \frac{a^2+b^2}{2} \\ P_{|\searrow\rangle} &= |d|^2 = \frac{(a-ib)(a+ib)}{2} = \frac{a^2+b^2}{2} \end{aligned} \right\} \text{iff } a, b \in \mathbb{R} \text{ (which is same)}$$

1.6 Case $a, b \in \mathbb{C}$

However if $a, b \in \mathbb{C}$, $\begin{pmatrix} a = x + iy \\ b = p + iq \end{pmatrix}$

$$P_{|\nearrow\rangle} = \frac{|a - ib|^2}{2} = \left| \frac{(x + iy - i(p + iq))}{\sqrt{2}} \right|^2 = \left| \frac{(x + q) + i(y - p)}{\sqrt{2}} \right|^2 = \frac{(x + q)^2 + (y - p)^2}{2}$$

$$P_{|\swarrow\rangle} = \frac{|a + ib|^2}{2} = \frac{(x - q)^2 + (y + p)^2}{2}$$

which are unequal iff $a, b \in \mathbb{C}$. (In general $a, b \in \mathbb{C}$).

2 Lec 22

|lec 22> is home assignment and will be covered at end of course.

3 Lec 23 : Addition of angular momenta

If we have two angular momenta J_1 and J_2 then

If J represents total angular momentum operator

$$\vec{J} = \vec{J}_1 + \vec{J}_2$$

then,

$$\left[|j_1 - j_2| \leq j \leq j_1 + j_2 \right] \quad \begin{array}{l} j = \text{total angular momentum} \\ \text{quantum numbers.} \end{array}$$

(for) e^- in d orbital ($l = 2$), ($s = 1/2$) for e^- , $(\vec{J} = \vec{L} + \vec{S})$

$$|l - s| \leq j \leq l + s$$

$$\frac{3}{2} \leq j \leq \frac{5}{2} \quad \text{in steps of one}$$

$$j = \frac{3}{2}, \frac{5}{2}$$

3.1 Total angular momentum states

The spin & orbital (total angular momentum states) are represented as $|j, m\rangle$

$$m : -j, -j + 1, \dots, j - 1, j$$

for $j = 3/2$, The possible states are

$$\left| \frac{3}{2}, -\frac{3}{2} \right\rangle, \left| \frac{3}{2}, -\frac{1}{2} \right\rangle, \left| \frac{3}{2}, \frac{1}{2} \right\rangle, \left| \frac{3}{2}, \frac{3}{2} \right\rangle \quad (4 \text{ states})$$

such that

$$J^2 |j, m\rangle = j(j + 1)\hbar^2 |j, m\rangle, \quad J_z |j, m\rangle = m\hbar |j, m\rangle$$

for each total angular momentum states.

Since $\vec{J} = \vec{L} + \vec{S}$,

$$J_z = \vec{L}_z + \vec{S}_z$$

where

$$\hat{L}_z |l, m_l\rangle = \hbar m_l |l, m_l\rangle, \quad L^2 |l, m_l\rangle = l(l+1)\hbar^2 |l, m_l\rangle$$

and,

$$\hat{S}_z |s, m_s\rangle = \hbar m_s |s, m_s\rangle, \quad S^2 |s, m_s\rangle = s(s+1)\hbar^2 |s, m_s\rangle$$

orbital angular states for $l = 2$:

$$|l, m_l\rangle = |2, -2\rangle, |2, -1\rangle, |2, 0\rangle, |2, 1\rangle, |2, 2\rangle$$

spin angular momentum states for e^- ($s = 1/2$):

$$|s, m_s\rangle = \left|\frac{1}{2}, -\frac{1}{2}\right\rangle, \left|\frac{1}{2}, +\frac{1}{2}\right\rangle$$

also for $j = 5/2$ the total angular momentum states are

$$|j, m\rangle = \left|\frac{5}{2}, -\frac{5}{2}\right\rangle, \left|\frac{5}{2}, -\frac{3}{2}\right\rangle, \left|\frac{5}{2}, -\frac{1}{2}\right\rangle, \left|\frac{5}{2}, \frac{1}{2}\right\rangle, \left|\frac{5}{2}, \frac{3}{2}\right\rangle, \left|\frac{5}{2}, \frac{5}{2}\right\rangle$$

i.e. 6 states

So for total of 4 + 6 states exist. (10 states), These 10 states came from individual states of $\hat{L}, \hat{S}$. ($5 \times 2 = 10$).

3.2 $l = 0$ case

If $l = 0$, (In s orbital state).

$$|l - s| \leq j \leq (l + s) \quad (j = 1/2)$$

so total angular momentum states are $|j, m\rangle, \left|\frac{1}{2}, -\frac{1}{2}\right\rangle$ and $\left|\frac{1}{2}, \frac{1}{2}\right\rangle$ i.e. only spin angular states.

$$J = \vec{L} + \vec{S}$$

$$J^2 |j, m\rangle = j(j+1)\hbar^2 |j, m\rangle \Rightarrow J^2 \left|\frac{1}{2}, \frac{1}{2}\right\rangle = \frac{1}{2} \left(\frac{1}{2} + 1\right) \hbar^2 \left|\frac{1}{2}, \frac{1}{2}\right\rangle$$

$$J_z |j, m\rangle = m\hbar |j, m\rangle \Rightarrow J_z \left|\frac{1}{2}, \frac{1}{2}\right\rangle = \frac{1}{2}\hbar \left|\frac{1}{2}, \frac{1}{2}\right\rangle$$

So, the total angular momentum states are in fact the spin angular momentum states.

3.3 States of H atom

The states of H atom are defined by quantum numbers.

$$|n, l, m_l; m_s\rangle$$

$$n = 1, 2, 3 \dots \infty$$

$$l = 0, 1, \dots n-1$$

$$m = -l \text{ to } +l$$

$$\text{for } s = 1/2$$

$$m_s = -1/2 \text{ to } +1/2$$

as, $E_{n,l,m,m_s} = f(n)$ alone.

$g \rightarrow$ degeneracy for a given ' n '

3.4 Degeneracy

$$\begin{aligned}
 g_n &= \sum_{l=0}^{n-1} \sum_{m_l=-l}^{+l} \underbrace{\sum_{m_s=-1/2}^{+1/2}}_2 (1) \\
 &= 2 \sum_{l=0}^{n-1} (2l+1) = 2 \left[2 \frac{n(n-1)}{2} + n \right] = 2[(n-1)n + n] = 2n^2 \\
 g_n &= 2 \sum_{l=0}^{n-1} (2l+1) = 2n^2 \quad \left| \begin{array}{l} \sum_{l=0}^{n-1} l = \frac{n(n-1)}{2} \\ \sum_{l=0}^{n-1} 1 = n \quad (n-1+1) \text{ terms} \end{array} \right.
 \end{aligned}$$

If we ignore $m_s = \pm 1/2$ then ($g_n = n^2$).

(side note) The energy levels only depend upon 'n' and not on any other quantum no. for $\frac{1}{r}$ potential unless we break the degeneracy by including some other interaction i.e. $\vec{B}_{\text{applied}}$ can couple with spin of e^- , and $\hat{H}_{\text{mag}} = -\vec{\mu}_e \cdot \vec{B}$ and that will distinguish b/w $s = +1/2$ and $s = -1/2$ state. The energy levels can depend on l, m_l, m_s etc.

$$\vec{\mu}_e = \frac{ge}{2m} \vec{S}$$

3.5 Spin-orbit coupling

If we switch on the $\vec{B}$ there will be coupling b/w internal intrinsic magnetic moment of e^- and the ext. $\vec{B}$, that will break this degeneracy in m_s . Then if we have anything / any potential which is not central then there is also dependence on m .

In general if we have central potential alone $V(r)$. $E = E(n, l)$. so for a given n, l ; m takes values $-l$ to $+l$ ($2l+1$) values

$$\begin{aligned}
 &\text{degeneracy including spin} = 2(2l+1) \\
 &\text{excluding spin} = 2l+1
 \end{aligned}$$

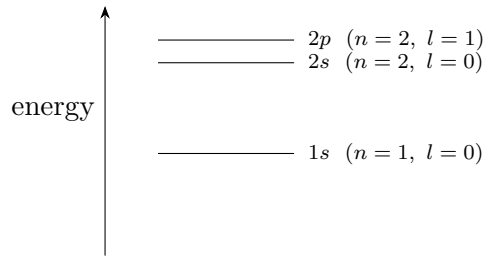
Note :- In practice even in the H-atom, there is spin-orbit coupling. i.e. coupling between spin of e^- and its orbital angular momentum; in the following sense (Its actually more complicated effect but its as if pretend for a minute that we are on its rest frame of e^- , we still have this intrinsic angular momentum ($\vec{S}$), we see proton (p^+) going around it; which is current loop and therefore there is dipole moment / magnetic moment associated with it (orbital motion) and that $\mu_e \propto \vec{L}$ and there will be term in $\hat{H}$ which is $\vec{L} \cdot \vec{S}$. (L - S coupling)

The moment we have L - S coupling we have broken the symmetry that this coulomb potential has, therefore the energy levels become dependent on l .

3.6 Spectrum of H atom

so energy for different l becomes different for same given 'n'.

And that is the reason spectrum of H-atoms looks like. (below)



$2p$ states and $2s$ states have different energy because of spin orbit coupling (without spin orbit coupling, $2s$ & $2p$ states will have same energy).

Similarly, $3s, 3p, 3d$ states are also split because of spin-orbit couplings.

The effective split is very small as compared to actual energy values of $2s$ or $2p$ states (very small splitting).

But as you go up the periodic table and as the (Z) atomic number increases, it turns out spin-orbit coupling becomes more and more prominent and there comes a stage when spin-orbit coupling effect is quite significant, in fact it is so significant and there are other effects as well (Hyperfine splitting which arises from coupling between magnetic moment of e^- ($\mu_e = -\mu_B \vec{\sigma}$) and magnetic moment of nucleus itself). Hyperfine splitting is also very small but as Z (atomic number) increases this effect become sufficiently large and we detect significant amount of splitting (and then we need ordering of these energy states ($s, p, d \dots$))

The spin-orbit coupling and ext $\vec{B}$, all serves to break the symmetry that the original H-atom has (coulomb potential).

3.7 Symmetry breaking

“So the general lesson is that the application of field breaks symmetry”.

After all the space in our room is isotropic, but all the directions are not equivalent as there is gravitation field in vertical dir. which breaks the translational symmetry. And that's the reason if you have particle moving around this room under effect of gravity; the ' x & y ' component of its momentum are conserved (in classical picture) as there is no force in X - Y direction, but p_z is certainly not conserved (because of gravity). So linear momentum conservation is broken in one direction because of the symmetry breaking field (gravity).

4 Problems

4.1 (Q) Delta function barrier

(Q) The particle of mass ' m ' moving in one dimension is incident upon a $\delta(x - x_0)$ barrier. Is the reflection coefficient is identically zero.

Ans :- No, that is not true, there is reflection as well as transmission.

4.2 (Q) Normalizable eigen state of $a^\dagger$

(Q) The only normalizable eigen state of the raising operator $a^\dagger$ of the linear harmonic oscillator is the ground state of hamiltonian. (Is this true or false)?

$$a^\dagger = \frac{x}{\sqrt{2m/\omega\hbar}} - \frac{ip}{\sqrt{2m\omega\hbar}}$$

$$a^\dagger |\lambda\rangle = \lambda |\lambda\rangle \quad \text{let } \lambda(x) = \langle x|\lambda\rangle \text{ be eigen state}$$

$$\begin{aligned}
\langle x | \frac{x}{\sqrt{2m/\omega\hbar}} + \frac{(-i\hbar\frac{\partial}{\partial x})}{\sqrt{2m\omega\hbar}} | \lambda \rangle &= \lambda \langle x | \lambda \rangle \\
\frac{x \langle x | \lambda \rangle}{\sqrt{2m/\omega\hbar}} + \frac{i^2\hbar}{\sqrt{2m\omega\hbar}} \frac{\partial}{\partial x} \lambda(x) &= \lambda \lambda(x) \Rightarrow \frac{x \lambda(x)}{2m} - \hbar \frac{\partial}{\partial x} \lambda(x) = \lambda \lambda(x) \\
\int \frac{\partial \lambda(x)}{\lambda(x)} &= \int \frac{1}{\hbar} \left(\frac{x}{2m} - \lambda \right) dx \\
\lambda(x) &\propto e^{\frac{1}{\hbar} \left(\frac{x^2}{4m} - \lambda x \right)}
\end{aligned}$$

$\lambda(x)$ has no-normalizable states so the statement is *false*. (but $\hat{a}$ has normalizable eigen states called as coherent states and also called as minimum uncertainty states)

4.3 (Q) Commutation relations and normalizability

(Q) [True / False.]

Angular momentum commutation relations together with the requirement that the eigen states of angular momentum be normalizable suffice to determine the possible eigen values of J^2 and $\vec{J} \cdot \hat{n}$ where $\hat{n}$ is unit vector along any arbitrary direction.

Ans True. (If we do not require normalizable eigen states then there is no guarantee that the spectrum of J^2 , $\vec{J} \cdot \hat{n}$ is discrete. we could have continuous spectrum for same operator).

In these cases, given an operator its spectrum is decided by what class of eigen states you would like to look at (use) for physical reasons. we want normalizability for conservation of probability and the interpretation of quantum mechanics. But mathematically there could be (more) continuous spectrum with non-normalizable states.

4.4 (Q) Symmetric potential and parity

(Q) [True / False].

A particle moves in 1-D in symmetric potential ($V(x) = V(-x)$) all the energy levels of the particle are discrete (given that we only have bound states) i.e. all wave functions dies down at $x = \pm\infty$. (exponentially faster). The position space wave functions of the particle in the ground state $\phi_0(x)$ is an even function of x .

Ans. True.

$$\begin{aligned}
i\hbar \frac{d}{dt} |\psi(t)\rangle &= \hat{H} |\psi(t)\rangle \\
|\psi(t)\rangle &= \sum_n c_n |\phi_n(t)\rangle \\
\langle x | \psi(t) \rangle &= \sum_n c_n \phi_n(x, t)
\end{aligned}$$

where,

$$\begin{aligned}
\phi_n(x, t) &= e^{-\frac{iE_n t}{\hbar}} \phi_n(x, 0) \\
-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \phi_n(x) + V(x) \phi_n(x) &= E_n \phi_n(x) \\
x' &= -x \\
-\frac{\hbar^2}{2m} \frac{\partial^2}{\partial x'^2} \phi_n(x) + V(-x) \phi_n(-x) &= E_n \phi_n(-x)
\end{aligned}$$

If $\phi(x)$ is solution to $H\phi = E\phi$ so is $\phi(-x)$ is a solution to $\hat{H}\phi = E\phi$

$\phi_n(-x)$ has to be some linearly dependent on $\phi(x)$ as there can't be another solution for same energy eigen values because of lack of degeneracy in (1-D).

$$\phi(-x) = c \phi(x)$$

$$\phi(-(-x)) = c(\phi(-x))$$

$$\phi(x) = c \phi(-x) = c(c \phi(x)) \Rightarrow c^2 = 1 \Rightarrow c = \pm 1$$

$$\phi(x) = \begin{cases} \phi(x) & \text{even fun} \\ -\phi(x) & \text{odd function} \end{cases}$$

“levels (energy levels) are non degenerate because we do not have any other operator which commutes with ‘ H ’ other than $\hat{H}$ itself”.

$[H, P] = 0 \Rightarrow$ There exists a common eigen set b/w $\hat{H}, \hat{P}$

Now parity operator $\hat{P}$ has eigen functions as all even functions and all odd functions, but $\hat{H}$ has special eigen functions. such that every eigen state of $\hat{H}$ is also a parity eigen state but converse is not true.

4.5 (Q) Oscillator eigen functions as basis

(Q) [True / False].

consider set of functions $\phi_n(x)$ (oscillator eigen fun. $\langle x|n\rangle$) any vector / function in x -basis can be written as some unique linear combination of $\phi_n(x)$.

True, as $\phi_n(x)$ makes orthonormal basis set.

$$f(x) = \sum c_n \phi_n(x) \quad \text{or} \quad \int_{-\infty}^{\infty} c_n \phi_n(x) dx$$

(A) If we consider $\phi_n(x)$ from 0 to ∞ to be a member of L_2 then e^{-x} would do the job (Legendre polynomials). and exactly that appears in radial wave function of H-atom.

4.6 (Q) Can $\Delta S_x \Delta S_y = 0$

(Q) Consider a particle with spin quantum no (s) = $\frac{1}{2}$, the particle can never be in the spin state $\Delta S_x \Delta S_y = 0$.

False.

$$\begin{aligned} (\Delta S_x)(\Delta S_y) &\geq \frac{1}{2} |\langle [S_x, S_y] \rangle| \quad [S_x, S_y] = i\hbar S_z \\ &\geq \frac{\hbar}{2} |\langle S_z \rangle| \cdots (|i| = 1) \quad |\langle [S_x, S_y] \rangle| = \hbar |\langle S_z \rangle| \quad (\text{as } |i| = 1) \end{aligned}$$

$$\langle S_z \rangle = \langle \psi | S_z | \psi \rangle, \quad S_z = \frac{\hbar}{2} \sigma_3 = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

$$|\psi\rangle_{spin} = \begin{pmatrix} a \\ b \end{pmatrix}$$

$$\langle S_z \rangle = (a^* \ b^*) \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \begin{pmatrix} a \\ b \end{pmatrix} \Rightarrow \frac{\hbar}{2} (a^* \ b^*) \begin{pmatrix} a \\ -b \end{pmatrix} = \frac{\hbar}{2} (|a|^2 - |b|^2)$$

$$\langle S_z \rangle = 0 \quad \text{if} \quad |a|^2 = |b|^2 \quad \sim (a = b).$$

so the statement is *false*.

4.7 (Q) Eigen functions of fourier transform operator

(Q) the functions (Harmonic oscillator functions) $\tilde{\phi}(p)$ are eigen functions of fourier transform operator.

True.

$$\begin{aligned}\tilde{\phi}(p) &= \langle p | \phi \rangle \\ &= \langle p | I | \phi \rangle = \langle p | \int | x \rangle \langle x | dx | \phi \rangle \\ &= \int_{-\infty}^{\infty} \langle p | x \rangle \langle x | \phi \rangle dx \quad \left(\langle p | x \rangle \propto e^{-ipx/\hbar} \right) \\ \tilde{\phi}(p) &= \int_{-\infty}^{\infty} e^{-\frac{ipx}{\hbar}} \phi(x) dx\end{aligned}$$

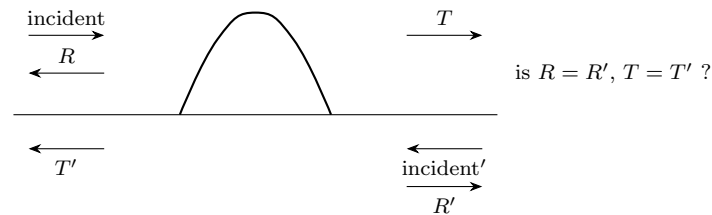
$\tilde{\phi}(p)$ is fourier transform of $\phi(x)$.

There also exists functions whose fourier transform is constant times another function, what can those eigen values be, fourth root of unity.

we know that fourth power of fourier operator is unity. so those eigen functions can be $1, i, -1, -i$. the ground state of oscillator corresponds to eigen value 1.

4.8 (Q) Is $R = R'$ and $T = T'$

(Q) consider the 1-D potential barrier $V(x)$, let the reflections and Transmission coefficients be R and T .



R' = The reflection coefficient for incident particle with (–ve momentum).

Ans, Yes $R' = R, T' = T$. (It does not require symmetry property for $V(x)$ (potential))

for any potential barrier $R' = R, T' = T$.

4.9 (Q) Expectation value of J^2

(Q) Let j be the total angular momentum quantum no of a system, then in any state $|\psi(t)\rangle$ of a system the operator J^2 must necessarily have the expectation value $\hbar^2 j(j+1)$.

Ans. True.

Since ‘ j ’ is specified so for any state $|\psi\rangle = |j, m\rangle$

$$J^2 |j, m\rangle = \hbar^2 j(j+1) |j, m\rangle$$

so

$$\langle \psi | J^2 | \psi \rangle = \hbar^2 j(j+1)$$

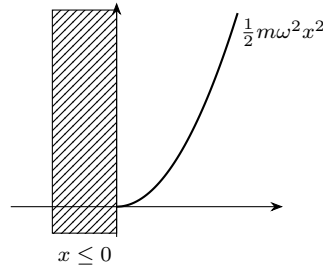
like wise $s = 1/2$ for e^-

$$\begin{aligned}S^2 \left| \frac{1}{2}, m_s \right\rangle &= \hbar^2 \left(\frac{1}{2} \right) \left(\frac{1}{2} + 1 \right) \left| \frac{1}{2}, m_s \right\rangle \\ \langle S^2 \rangle &= \left\langle \frac{1}{2}, m_s \right| S^2 \left| \frac{1}{2}, m_s \right\rangle = \frac{3}{4} \hbar^2\end{aligned}$$

4.10 (Q) Half oscillator potential

(Q) A particle moves in a half oscillator potential,

$$V(x) = \begin{cases} \frac{1}{2}m\omega^2 x^2 & x > 0 \\ \infty & x \leq 0 \end{cases}$$



Find ground state energy for a particle of mass 'm'.

we have, for $x > 0$

$$-\frac{\hbar^2}{2m} \frac{\partial^2 \phi}{\partial x^2} + \frac{1}{2}m\omega^2 x^2 \phi = E\phi \quad \text{①}$$

for $x \leq 0$

$$\phi = 0.$$

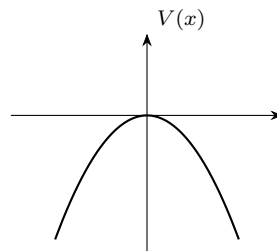
when we solve equation ① and we impose condition of square-integrability (function to be finite at ∞) The energy levels turn out to be discrete, solutions will again be Hermite polynomials, but solution will be those Hermite polynomials which vanish at origin. (odd ones $n = 1, 3, 5 \dots$).

$$E = \left(n + \frac{1}{2}\right) \hbar\omega \quad \left(\begin{array}{l} n = 2, 4, 6 \text{ are not} \\ \text{allowed} \end{array} \right)$$

$$E_{g.s} = E_1 = \frac{3}{2}\hbar\omega \quad (\text{ground state}).$$

and $H_1, H_3, H_5 \dots$ now forms a complete set. and any vector now can be expanded in $\{H_1, H_3, H_5 \dots \text{basis}\}$.

what if $V(x) = -\frac{1}{2}m\omega^2 x^2$



$\left\{ \begin{array}{l} \text{No bound states exists} \\ \text{No normalizable states exists.} \end{array} \right.$

(side note) Anything which is periodic in classical mechanics becomes bound state in QM.

4.11 (Q) Perturbed (shifted) oscillator

(Q) The hamiltonian of perturbed oscillator is given by

$$\hat{H} = \hbar\omega \left(a^\dagger a + \frac{1}{2} \right) + \lambda \left(a + a^\dagger \right)$$

what is the value of ground state energy eigen value.

$$a = \frac{x}{\sqrt{2m/\omega\hbar}} + \frac{ip}{\sqrt{2m\omega\hbar}}, \quad a^\dagger = \frac{x}{\sqrt{2m/\omega\hbar}} - \frac{ip}{\sqrt{2m\omega\hbar}}$$

$$a + a^\dagger = \frac{2x}{\sqrt{2m/\omega\hbar}} \quad \text{so it is a shifted oscillator}$$

$$H = \hbar\omega \left[a^\dagger a + \frac{\lambda}{\hbar\omega} \left(a + a^\dagger \right) \right] + \frac{1}{2}\hbar\omega$$

let us define : $b = a + \frac{\lambda}{\hbar\omega}$

so, $b^\dagger = a^\dagger + \frac{\lambda}{\hbar\omega}$, $[b, b^\dagger] = 1$

$$\hat{H} = \hbar\omega \left(\left(b^\dagger - \frac{\lambda}{\hbar\omega} \right) \left(b - \frac{\lambda}{\hbar\omega} \right) + \frac{\lambda a}{\hbar\omega} + \frac{\lambda a^\dagger}{\hbar\omega} \right) + \frac{1}{2}\hbar\omega$$

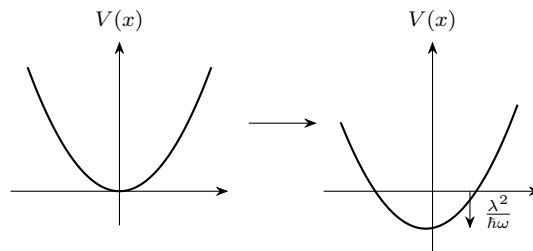
$$\hat{H} = \hbar\omega \left(b^\dagger b + \frac{\lambda^2}{(\hbar\omega)^2} - \frac{(b^\dagger + b)\lambda}{\hbar\omega} + \frac{\lambda}{\hbar\omega} \left(b + b^\dagger - \frac{2\lambda}{\hbar\omega} \right) \right) + \frac{1}{2}\hbar\omega$$

$$\hat{H} = \hbar\omega \left(b^\dagger b - \frac{\lambda^2}{\hbar^2\omega^2} \right) + \frac{1}{2}\hbar\omega$$

$$E = \frac{\hbar\omega}{2} + \left(n - \frac{\lambda^2}{\hbar^2\omega^2} \right) \hbar\omega$$

$$E = \hbar\omega \left(n - \frac{\lambda^2}{\omega^2\hbar^2} + \frac{1}{2} \right)$$

$$E_{\text{ground}} = \hbar\omega \left(\frac{1}{2} - \frac{\lambda^2}{\omega^2\hbar^2} \right) = \frac{\hbar\omega}{2} - \frac{\lambda^2}{\hbar\omega}$$



4.12 (Q) Commutator of $\vec{S} \cdot \hat{n}$ and $\vec{S} \cdot \hat{n}'$

(Q) Let $\vec{S}$ be the spin operator for spin 1/2 particle, and $\hat{n}, \hat{n}'$ are two arbitrary directions find $[\vec{S} \cdot \hat{n}, \vec{S} \cdot \hat{n}']$.

$$\begin{aligned} [\vec{S} \cdot \hat{n}, \vec{S} \cdot \hat{n}'] &= \frac{\hbar^2}{4} [\vec{\sigma} \cdot \hat{n}, \vec{\sigma} \cdot \hat{n}'] \\ \left(\vec{S} \cdot \hat{n} = \frac{\hbar}{2} \vec{\sigma} \cdot \hat{n} \right) &= \frac{i\hbar^2}{2} (\hat{n} \times \hat{n}') \cdot \vec{\sigma} \end{aligned}$$

4.13 (Q) $S_x S_y$ in up-down basis

(Q) For $|\uparrow\rangle$ and $|\downarrow\rangle$ usual eigen states, what is $S_x S_y$? in $|\uparrow\rangle$ and $|\downarrow\rangle$ basis.

$$\begin{aligned} S_x S_y &= \frac{\hbar^2}{4} \sigma_x \sigma_y = \frac{\hbar^2}{4} (i\sigma_z) \\ &= \frac{i\hbar}{2} \left(\frac{\hbar}{2} \sigma_z \right) = \frac{i\hbar}{2} S_z \\ S_z &= \frac{\hbar}{2} \sigma_3 = \frac{\hbar}{2} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ S_x S_y &= \frac{i\hbar^2}{4} \sigma_3 = \frac{i\hbar^2}{4} \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} \\ S_x S_y &= \frac{i\hbar^2}{4} \begin{bmatrix} |\uparrow\rangle \langle\uparrow| & -|\downarrow\rangle \langle\downarrow| \end{bmatrix} \end{aligned}$$

4.14 (Q) Expectation value of $S_x + iS_y$

(Q) In the state $\begin{pmatrix} \cos \theta/2 \\ e^{i\phi} \sin \theta/2 \end{pmatrix}$ find $\langle S_x + iS_y \rangle$.

$$\begin{aligned} S_x + iS_y &= \left(\frac{\hbar}{2} \sigma_x + i \frac{\hbar}{2} \sigma_y \right) \\ \sigma_x &= \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \\ S_x + iS_y &= \frac{\hbar}{2} \begin{pmatrix} 0 & 2 \\ 0 & 0 \end{pmatrix} = \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \\ \langle S_x + iS_y \rangle &= \langle \psi | S_x + iS_y | \psi \rangle \\ &= \begin{pmatrix} \cos \frac{\theta}{2} & e^{+i\phi} \sin \frac{\theta}{2} \end{pmatrix} \hbar \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix} \begin{pmatrix} \cos \theta/2 \\ e^{-i\phi} \sin \theta/2 \end{pmatrix} \\ &= \hbar \begin{pmatrix} \cos \frac{\theta}{2} & e^{i\phi} \sin \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} e^{-i\phi} \sin \frac{\theta}{2} \\ 0 \end{pmatrix} \\ \langle S_x + iS_y \rangle &= \frac{\hbar}{2} \left(e^{-i\phi} \sin \theta \right) \end{aligned}$$

notice that at $\theta = 0$, $\langle S_x + iS_y \rangle = 0$

For $\theta = 0$, the eigenstate $\begin{pmatrix} 1 \\ 0 \end{pmatrix}$ is eigen state of S_z itself.

then $\langle S_x \rangle$ and $\langle S_y \rangle$ both vanishes.

4.15 (Q) Shrodinger eqn for free particle

(Q) write shrodinger eqn for a free particle. (Find propagator ?)

$$\begin{aligned} e^{a \frac{d}{dx}} f(x) &= f(x+a) \\ e^{b \frac{d^2}{dx^2}} f(x) &= \int_{-\infty}^{\infty} dx' K(x, x') f(x') \quad \left(K \sim e^{-(x-x')^2} \right) \end{aligned}$$



4.16 (Q) $J_+ |j, m\rangle$

(Q) find $J_+ |j, m\rangle$

$$J_+ |j, m\rangle = \sqrt{(j-m)(j+m+1)} |j, m+1\rangle$$

$$J_- |j, m\rangle = \sqrt{(j+m)(j-m+1)} |j, m-1\rangle$$

4.17 (Q) E.O.M. satisfied by $\langle x \rangle$

(Q) E.O.M. satisfied by $\langle x \rangle$ of particle, when it is in eigen state of p momentum operator with eigen value p is ?

$$\frac{d}{dt} \langle \hat{x} \rangle = \frac{\langle \hat{p} \rangle}{m}$$

4.18 (Q) Expectation value of $a^\dagger a$ in coherent state

(Q) The expectation value of $a^\dagger a$ in coherent state $|\alpha\rangle$ of linear harmonic oscillator is :-

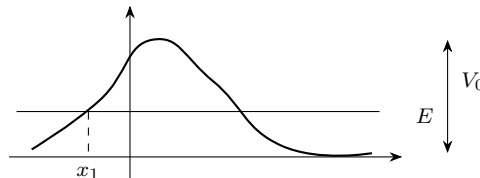
$$|\alpha\rangle = e^{-\frac{1}{2}|\alpha|^2} \sum_{n=0}^{\infty} \frac{\alpha^n}{\sqrt{n!}} |n\rangle$$

$$\begin{aligned} \langle a^\dagger a \rangle &= \langle \alpha | a^\dagger a | \alpha \rangle \\ &= e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{\alpha^n \alpha^{*n}}{n!} \langle n | N | n \rangle \quad (\langle n | n \rangle = 1) \\ &= e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{\alpha^{*n} \alpha^n n}{n!} \\ &= e^{-|\alpha|^2} \sum_{n=0}^{\infty} \frac{(|\alpha|^2)^n}{n!} n = e^{-|\alpha|^2} |\alpha|^2 \sum_{n=1}^{\infty} \frac{(|\alpha|^2)^{n-1}}{(n-1)!} = |\alpha|^2 e^{-|\alpha|^2} e^{|\alpha|^2} \\ &= |\alpha|^2 \end{aligned}$$

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5.1 (Q) Barrier penetration

(Q) classically, if we have a potential barrier of height V_0 and if the total energy is $E < V_0$ then the particle crawls up and rolls back but does not climb further



It can not climb further and exists in $x_0 > x_1$ because if it climb further the K.E < 0 .
so Is the same reason valid for quantum mechanical particles.

Ans No, Quantum mechanically the particle penetrates in barrier and wavefunction in barrier dies down exponentially (e^{-kx}) ($k = \frac{\sqrt{2m(V-E)}}{\hbar}$). When the particle is found inside the barrier $x > x_1$ is the K.E < 0 ; No $\langle p^2 \rangle$ has to be positive i.e. $\langle \psi | \hat{p} \hat{p} | \psi \rangle = \|\hat{p}\psi\|^2$ has to be positive $\forall \psi$. Moreover, we can't ask when particle is at $x = x_1$ what is its K.E as there is uncertainty in momentum or we can say that $[H, \frac{p^2}{2m}] \neq 0$ or $[\frac{p^2}{2m}, x] \neq 0$ so we do not have common eigen states.

So the question is not a valid one.

5.2 (Q) Nature of spectra of $\hat{x}$ and $\hat{p}$

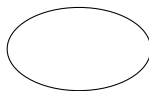
(Q) what is the nature of spectra of $\hat{x}$ and $\hat{p}$.

classically $\left. \begin{array}{l} -\infty < x < \infty \\ -\infty < p < \infty \end{array} \right\}$ continuous spectrum for $\hat{x}$ and $\hat{p}$.

So quantum mechanically, we can have all eigen values of $\hat{x}$ and $\hat{p}$. so the spectra is also continuous.

If we put particle / system in boundary condition / box. Then the wavefunction must vanish at boundary points, and this can lead to discretization of spectra. **ex** eigen values of $\hat{H}$ i.e. E get quantized.

Even if particle does not have any boundary but it lives in a circle, then the eigen values of $\hat{p}$ gets quantized.



also L_z gets quantized.

$$-i\hbar \frac{\partial}{\partial \phi} F(\phi) = m\hbar F(\phi)$$

It turns out ($L_z : m\hbar$, $m = \text{integer}$.) if we require $F(\phi) = F(\phi + 2\pi)$.

$$\begin{aligned} F(\phi) &= A e^{im\phi} = A e^{im(\phi+2\pi)} \\ \Rightarrow A e^{im\phi} (1 - e^{im2\pi}) &= 0 \\ \Rightarrow (m2\pi) &= 2n\pi \quad n = 0, 1, 2 \dots \end{aligned}$$

'm' must be integer! ($m = n$)

5.3 (Q) Idea of spin

(Q) Idea of spin :- (Historical origin of spin)

spectra of atoms, as you know from various other courses. The spectral lines emitted by atoms coorespond to transition between various energy states available for e^- in atoms.

Long ago, when Q.M was first formulated and H-atom spectrum for instance was being explained in quantum mechanics It turned out that there were discrepancies between predictions of the normal Shrodinger equation for these spectral lines and what was actually observed (Results / observation). So various resolutions were proposed but the one that turned out to be the right / correct one had to do with idea of intrinsic angular momentum / spin; and this was postulated by various people perticularly by George Uhlenbeck and Samuel Goudsmit. They specifically said that there is some thing called spin and is two valued variable and then the famous experiment, Stern-Gerlach experiment.

5.4 Stern-Gerlach experiment

(We know that

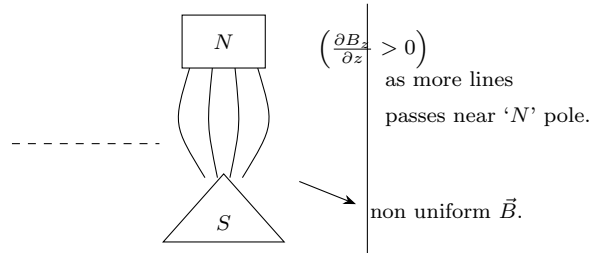
$$\begin{aligned}\vec{\mu}_e &= \frac{ge}{2m_e} \vec{S}_e \\ &= \frac{-g|e|\hbar}{2m_e} \frac{\vec{\sigma}}{2} \\ &= - \left(\frac{|e|\hbar}{2m_e} \right) \vec{\sigma} \quad \longrightarrow \quad \mu_B \text{ (Bohr magneton)}\end{aligned}$$

How we measure $\vec{\mu}_e$; That is what shown in this experiment that μ_e has real measurable effect once we place these e^- in $\vec{B}$. But of course placing free electrons in $\vec{B}$ was tricky, so what they did was to take silver atoms ($47 e^-$ out of which $46 e^-$ contribute nothing to $\vec{B}$). and this whole shell of $46 e^-$ is spherically symmetric so we might assume that In ground state of Ag atom these $46 e^-$ are in state of total angular momentum zero state.

The 47^{th} electron is also in $l = 0$ state ($L = 0$) but it has the spin. This heavy particle now (the full Ag atom) essentially acts like single magnetic moment due to $47^{th} e^-$. (due to the intrinsic magnetic moment of e^-)

Now once we have magnetic moment and place it in $\vec{B}$, then there is $(-\vec{\mu} \cdot \vec{B})$ potential. and if $\vec{B}$ is inhomogeneous (non-uniform) there is a force which is ($F = -\nabla U$)

So the idea they had was to prepare a beam of silver ions (monochromatic beam) in the sense that its collimated, mono energetic ...)



$$F \sim \frac{\partial}{\partial z} (\mu B_z) = \mu_e \frac{\partial B_z}{\partial z}$$

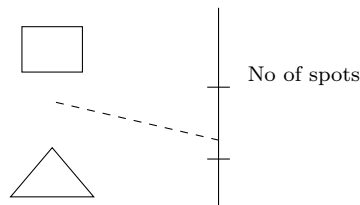
therefore if B_z changes along z direction significantly we have

$$F = \pm \mu_e \frac{\partial B_z}{\partial z}$$

which can be different forces depending upon μ_e which can have positive as well as negative eigen value. ($\mu_e \propto \vec{\sigma}$).

In one case Force will be upwards and in other case force will be downwards.

self interpreted $\left(\begin{array}{l} \text{Force is in } +z \text{ direction if } \frac{\partial B_z}{\partial z} > 0 \text{ and } \sigma_z = +\frac{1}{2} \\ \text{For spin } (+1/2) e^- \text{ the direction of force is downwards} \end{array} \right.$



depending on it we can determine how many spin states it has.

for e^- it turns out only two states exist ($|\frac{1}{2}, \frac{1}{2}\rangle$ and $|\frac{1}{2}, -\frac{1}{2}\rangle$)

From this experiment; since only two states exists $s = 1/2$ for e^- ; if 4 spots existed in place of two then the spin quantum no (s) would have been $(3/2)$.

In H-atom itself we have spin orbit coupling. where we have extra $H_{cop} = \vec{L} \cdot \vec{S}$ which breaks coulombs central potential and Energy levels starts depending on ' l '. Thus removing the degeneracy.

Note that, If the outermost e^- were in $l = 2$ (d state) then total angular momentum would have ' j ' total angular quantum no.

$$|j - s| \leq j \leq l + s \quad \begin{array}{l} s = 1/2 \\ l = 2 \end{array}$$